BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 2 Practical No: 1

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Subject Name : Deep Learning

Problem Statement

Install and configure TensorFlow/Keras and perform basic preprocessing and visualization of the MNIST handwritten digit dataset.

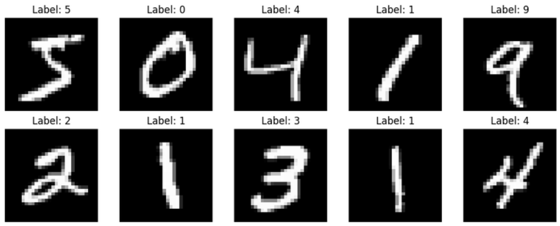
Algorithm

1. Begin.
2. Install TensorFlow and access the Keras deep learning API.
3. Import TensorFlow, Keras, NumPy, and Matplotlib libraries.
4. Display the installed TensorFlow and Keras versions.
5. Load the MNIST handwritten digit dataset using Keras.
6. Separate the dataset into training images, training labels, testing images, and testing labels.
7. Display the dimensions of the training and testing datasets.
8. Check the data type and original pixel value range.
9. Visualize sample handwritten digit images with their corresponding labels.
10. Convert image data to floating-point format.
11. Normalize pixel values from the range 0–255 to the range 0–1.
12. Verify the normalized pixel value range.
13. Visualize sample images after normalization.
14. Display the number of training and testing samples.
15. Display the shape of one input image.
16. End.

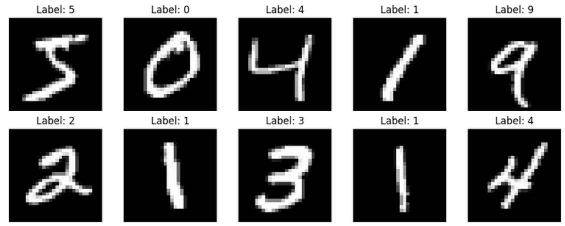
OUTPUT:

MNIST Dataset – Output

Sample Images Before Normalization



Sample Images After Normalization



Dataset and Preprocessing Results

Training images : 60,000
Testing images : 10,000
Image shape : 28 × 28
Data type : uint8
Original pixel range : 0 – 255
Normalized train range : 0.0 – 1.0
Normalized test range : 0.0 – 1.0

Dataset: MNIST handwritten digit images